

EMPIRICAL AIED & LEARNING ANALYTICS RESEARCH

Quantifying AI Over-Reliance in Programming Education: A Telemetry-Driven Behavioral Metric with Productive Friction in S-SPARC

An Empirical Framework for Diagnosing Epistemic Dependence, Latent Construct Modeling, and Closed-Loop Scaffolding on the E-STRANGE Learning Platform

Core Research Thesis

Generative AI generates executable code in seconds, but unconstrained access fosters severe cognitive offloading and epistemic dependence. This research formulates a real-time, telemetry-driven composite metric: the **AI Over-Reliance Index (ORI)**. Through empirical validation, latent factor modeling (EFA/CFA), and structured productive friction in S-SPARC, this study establishes a defensible paradigm to diagnose, quantify, and remediate cognitive atrophy in higher computer science education.

INSTITUTION & FACULTY Maranatha Christian University Faculty of Information Technology	RESEARCH DOMAINS AI in Education (AIED) Learning Analytics & EDM	OPERATIONAL TESTBED S-SPARC AI Engine E-STRANGE Auto-Grader	PUBLICATION TARGETS IEEE TALE / EDUCON / SIGCSE Scopus Q1 (IEEE TLT / C&E: AI)
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Cognitive Atrophy & Epistemic Dependence in Programming Labs

Problem Formulation

CLASSROOM REALITY

The Passive Behavioral Feedback Loop

Students routinely bypass mental reasoning through an automated copy-paste loop:

Copy Problem Statement → Prompt AI → Copy Generated Code → Submit to Auto-Grader

This cycle produces artificial success on automated unit test cases while masking critical cognitive deficits that lead to severe failure during unassisted, closed-book evaluations.

Result: High lab assignment scores mask complete inability to code independently.

EMPIRICAL LITERATURE

Evidence from Peer-Reviewed Research

- **Critical Thinking Degradation:** Systematic review shows excessive reliance on conversational AI impairs analytical reasoning, problem decomposition, and self-debugging [2].
- **Lack of Cognitive Transfer:** Instant AI code generation exhibits near-zero correlation with independent conceptual mastery [3], [14].
- **Negative Final Grade Impact:** High frequency of LLM dependence correlates negatively with summative exam scores [4].
- **Surface-Level Regulation:** Analysis of 2,376 ChatGPT interactions proves students rarely engage in metacognitive reflection [5].

Literature Consensus: AI assistance without structured friction damages skill acquisition.

Pedagogical Diagnosis: The Mechanism of Cognitive Atrophy

Cognitive offloading eliminates the essential struggle of mental trace execution, edge-case debugging, and syntax structuring. When learners no longer grapple with syntax and logic errors independently, their foundational mental schema for algorithmic problem-solving atrophies [2], [6].

Research Gap: Why Current Solutions Fall Short

Comparative Analysis

Current Paradigms (2022-2026)	Inherent Methodological Limitations	S-SPARC Novel Research Approach
Subjective Surveys [2], [7]	Likert scales suffer from recall bias and social desirability; cannot capture real-time typing dynamics during problem-solving.	Continuous Telemetry Analytics: Millisecond-level event logging (keystroke cadence, reflection pauses, prompt entropy) with zero survey overhead.
Binary Plagiarism Detectors (AI Text Classifiers)	High false-positive rates; purely punitive without providing diagnostic feedback on student problem-solving strategies.	Cognitive Inquiry Profiling: Longitudinally maps problem decomposition, prompt sophistication, and epistemic autonomy [5].
Post-Hoc Lab / Eye-Tracking [15], [16]	Rigorous but confined to artificial lab environments; cannot provide a computable metric for live classroom intervention.	Real-Time Composite Index (ORI): Computes live over-reliance scores to trigger dynamic, adaptive pedagogical scaffolding in real time.
Frictionless Commercial AI [5], [8]	Commercial chatbots encourage rapid prompt spamming and full-code generation without enforcing metacognitive reflection.	Tiered Productive Friction: Enforces mandatory 60-second reflection pauses and multi-stage Socratic guidance [9], [10].

Scientific Synthesis of Research Gap

The academic literature currently lacks a unified framework that combines unobtrusive telemetry, information-theoretic prompt evaluation, and closed-loop productive friction directly integrated into an institutional LMS auto-grader.

AI Over-Reliance Index (ORI): Mathematical Formulation

Mathematical Model

$$ORI = 1 - (w1 \cdot Q_prompt + w2 \cdot S_CIOE + w3 \cdot R_concept + w4 \cdot F_reflect)$$

All component indicators normalized to [0, 1] • Total weight constraint: $w1 + w2 + w3 + w4 = 1.0$

INDICATOR 1**Q_prompt (w1 = 0.30)****Shannon Information Entropy**

$H(X) = -\sum p(x) \log_2 p(x)$ [11]. Measures lexical richness and technical syntax token density within coding prompts [13], [17].

Penalizes generic, low-effort queries like "fix code".

INDICATOR 2**S_CIOE (w2 = 0.25)****Context Completeness**

Measures compliance across 4 structured dimensions: Context, Input specs, Expected Output, and Error Trace [12].

Rewards rigorous problem specification.

INDICATOR 3**R_concept (w3 = 0.20)****Conceptual Ratio**

Ratio of conceptual/logic inquiries versus requests for direct code generation (Bloom Levels C2-C4 vs C1).

Differentiates learning from direct outsourcing.

INDICATOR 4**F_reflect (w4 = 0.25)****Friction Adherence**

Adherence to the 60s reflection window [9] coupled with keystroke cadence analysis to detect external copy-paste.

Prevents meta-gaming via external LLMs.

Psychometric Rigor Disclosure: The component weights above represent an exploratory baseline. Structural dimensionality and final weights are empirically calibrated via Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) across distinct student cohorts.

Four Pillars of Scientific Innovation

Original Contributions

CONTRIBUTION I

Real-Time Composite Metric for Adaptive Scaffolding

Unlike post-hoc reliance taxonomies [15] or eye-tracking setups [16] analyzed after lab sessions, S-SPARC computes ORI scores dynamically during active coding. This enables the platform to throttle AI assistance dynamically and activate Socratic scaffolding when over-reliance is detected.

Impact: First real-time pedagogical intervention engine for Generative AI in CS.

CONTRIBUTION II

Information-Theoretic Evaluation of Coding Prompts

Adapts Shannon Information Entropy $H(X)$ [11] to quantify lexical diversity, structural complexity, and technical token density in student programming prompts, advancing significantly beyond naive character-count or word-length metrics [13], [17].

Impact: Objective mathematical quantification of prompt engineering depth.

CONTRIBUTION III

Empirical Productive Friction & Anti Meta-Gaming

Empirically tests the pedagogical efficacy of mandatory 60-second reflection pauses [9], [10] reinforced by keystroke dwell-time analytics to verify human typing cadence and prevent students from bypassing the intervention via external LLMs.

Impact: Enforces authentic cognitive engagement rather than superficial compliance.

CONTRIBUTION IV

Closed-Loop AI Tutor & LMS Auto-Grader Integration

Directly correlates continuous AI conversational telemetry with deterministic unit-test outcomes on E-STRANGE (pass rates, execution runtime, debugging iterations) and unassisted exam performance to establish learning transfer.

Impact: Connects AI dialog quality directly to measurable code execution rigor.

Active Telemetry Pipeline (Zero-Friction Collection)

Operational Testbed

INFRASTRUCTURE LIVE

Operational System Readiness

- **Active Backend Logging:** Relational MySQL tables in S-SPARC and E-STRANGE record student interaction telemetry automatically with zero manual configuration.
- **Zero Operational Overhead:** Telemetry capture operates passively in the background without interrupting lab sessions or inducing survey fatigue.
- **Automated Export Pipeline:** Pre-built ETL scripts extract, sanitize, and structure raw data directly into Pandas DataFrames for statistical modeling.

Effort for Instructors: ZERO manual data entry or paper surveys required.

TELEMETRY SCHEMA

Telemetry Field	Data Definition & Granularity
prompt_raw_text	Raw query string and millisecond timestamp
entropy_score	Shannon Information Entropy H(X) score
cioe_compliance	4-factor boolean vector (Context, Input, Output, Error)
keystroke_dwell	Typing cadence & anti-paste verification metrics
execution_metrics	Unit test pass rate, runtime errors, attempt cycles
compute_footprint	Inference token consumption & estimated CO2 (g)

Data Privacy & Mitigation of Pygmalion / Labeling Bias

PILLAR 1

Informed Consent & Opt-Out

Students receive full transparency regarding aggregated data analytics for curriculum improvement, with guaranteed penalty-free opt-out options that do not affect academic grading.

PILLAR 2

De-Identification & Encryption

All personal identifiable information (Student ID, Name) is cryptographically hashed into unique random research identifiers prior to analysis. Datasets are stored on secure, access-controlled institutional servers.

PILLAR 3

Preventing Pygmalion Effect

Individual ORI scores and behavioral personas are strictly blind-masked from course instructors during the semester to prevent algorithmic labeling from biasing subjective grading or teaching behavior.

Instructor Dashboard Harmonization: Phase-Gated Deployment

The live instructor evaluation dashboard marketed in the S-SPARC product roadmap follows a phase-gated deployment: during Phase 1 (Validation Pilot), individual metrics remain blind-masked to protect internal validity. Full live dashboard access is unlocked in Phase 2 once psychometric and construct validity are rigorously established.

Mitigating Reactivity, Confounders, and Estimation Biases

Methodological Rigor

MITIGATION 1

Reactivity & Maturation Confound Mitigation

- **A-B Phase Design:** Modules 1-2 serve as an unconstrained baseline phase to record natural traits; Modules 3-4 introduce active productive friction (60s pause).
- **Confound Disclosure:** For a single-cohort design, maturation and friction effects are formally disclosed as a limitation; if ≥ 2 parallel sections are available, counterbalancing (A-B vs B-A) is implemented.

MITIGATION 2

Two-Phase Psychometric Validation Roadmap

- **Phase 1 (Semester 1 / Cohort 1):** Instrument development and Exploratory Factor Analysis (EFA) to discover latent factor loadings across the 4 ORI indicators.
- **Phase 2 (Semester 2 / Cohort 2):** Independent Confirmatory Factor Analysis (CFA) to verify construct unidimensionality and prevent model overfitting.

MITIGATION 3

Entropy Inversion Paradox (Interaction Modeling)

- **Interaction Model:** Formulates an *Entropy* $\times$ *Baseline Prior Ability* (pre-test / GPA) interaction term to ensure expert prompt conciseness is not misclassified as novice over-reliance.

MITIGATION 4

NLP Classifier Sub-Study Validation (R_concept)

- **Double Annotation Protocol:** Validates intent classification against ≥ 100 expert-annotated prompts by 2 independent raters, requiring Cohen's Kappa $\kappa \geq 0.80$ before formula deployment.

Anti-Circularity Validation, Cross-Validation, & Pre-Registration

PROTOCOL 1

Anti-Circularity Cross-Validation

Eliminates circular validation by strictly separating regression weight calibration from validation testing using a rigorous **5-Fold Cross-Validation** protocol.

PROTOCOL 2

Standardized Convergent Validity

Cross-validates ORI scores against an internationally validated instrument: the *GenAI Reliance Scale* (Hou et al., Computers & Education 2025 [1]).

PROTOCOL 3

OSF Pre-Registration Protocol

Locks the confirmatory analysis plan on the Open Science Framework (OSF.io) prior to data collection, eliminating p-hacking and researcher degrees of freedom.

Multicollinearity Diagnostics (Variance Inflation Factor • $VIF < 5.0$)

All predictor variables in the ORI formulation are evaluated for multicollinearity to ensure component orthogonality, parameter stability, and unambiguous interpretability of calibrated regression weights.

Four Exploratory Hypotheses with Rigorous Statistical Controls

Bonferroni Adjusted $\alpha = 0.0125$

HYPOTHESIS 1

H1. Volume-Driven Green AI Paradox

Hypothesis: Over-reliant students generate higher cumulative carbon emissions due to excessive prompt iteration volumes rather than per-query complexity.

Control: Controls for Module Difficulty Fixed-Effects and isolates query volume from semantic cache hit rates.

HYPOTHESIS 2

H2. AST Code Distance Delta vs Variable Renaming

Hypothesis: Autonomous students exhibit significantly larger Abstract Syntax Tree (AST) structural distances between AI suggestions and final submissions.

Control: Evaluates the final AI-suggested code block within the session against the student's submitted code.

HYPOTHESIS 3

H3. Reflexive Pause vs First-Attempt Pass Rate

Hypothesis: Active adherence to reflection pauses positively predicts first-attempt pass rates on auto-graded programming assignments.

Control: Incorporates baseline prior ability as a covariate; claims remain strictly associative rather than deterministic.

HYPOTHESIS 4

H4. Query Novelty & Semantic Cache Lifecycle

Hypothesis: Semantic cache misses reflect deeper student exploratory inquiry into complex edge cases and non-standard problem boundaries.

Control: Controls for cache maturity lifecycle and assignment timing to prevent cold-start artifacts.

Joint Publication Targets, Co-Authorship, & Generalization Limits

Deliverables & Scope

PUBLICATION PIPELINE

Target Publication Venues & Authorship Model

- **Phase 1: Top-Tier Peer-Reviewed International Conferences:**
 - IEEE International Conference on Teaching, Assessment, and Learning for Engineering (IEEE TALE)
 - IEEE Global Engineering Education Conference (IEEE EDUCON) / ACM SIGCSE
- **Phase 2: Scopus Q1 / High-Impact International Journals:**
 - IEEE Transactions on Learning Technologies (IEEE TLT)
 - Computers and Education: Artificial Intelligence (Elsevier)
- **Authorship Structure:** Academic Advisor as Principal Investigator / Corresponding Author alongside Student Research Team.

METHODOLOGICAL SCOPE

Boundary of Generalization

The ORI metric is specifically formulated for code-based problem-solving environments evaluable via **deterministic unit testing** within an automated grading framework.

Generalization to non-computing disciplines (e.g., qualitative essay composition or social sciences) requires non-deterministic evaluation rubrics and is formally defined as future work.

Two-Phase Research Execution Roadmap

PHASE 1 (SEMESTER 1)

PHASE 1: Instrument Development & Exploratory EFA (Current Semester • 8 Weeks)

Timeline	Primary Research Activities	Key Research Deliverables
Weeks 1 - 2	OSF Pre-Registration, Institutional Ethics Approval, Baseline Phase (Modules 1-2 Unconstrained)	Locked OSF protocol, baseline natural interaction telemetry
Weeks 3 - 4	Intervention Phase (Modules 3-4 with Active 60s Productive Friction)	Complete longitudinal pre/post intervention telemetry dataset
Weeks 5 - 6	MySQL ETL extraction, NLP double annotation ($\kappa \geq 0.80$), Exploratory Factor Analysis (EFA)	Psychometric factor report & calibrated regression baseline
Weeks 7 - 8	Manuscript drafting and submission to IEEE International Conference	Completed manuscript submitted to IEEE TALE / EDUCON

PHASE 2: Confirmatory Validation, Independent Replication, & Scopus Q1 Journal (Next Semester)

Independent cohort replication ($N \geq 80$), Confirmatory Factor Analysis (CFA), convergent validation against the standardized Hou et al. scale [1], testing the Entropy $\times$ Ability interaction model, and full manuscript submission to Scopus Q1 journals (IEEE TLT / C&E: AI) accompanied by the live instructor dashboard deployment.

Advisor Endorsement & Collaborative Action Items

Call to Action

ACTION ITEM 1

Practicum Class Deployment

Approval to deploy S-SPARC and E-STRANGE across 1 active programming lab cohort (35-50 students) during 4 practical lab modules.

Zero disruption to established course syllabus.

ACTION ITEM 2

Ethics & OSF Endorsement

Institutional endorsement for the faculty ethics review and co-sponsorship of the OSF.io pre-registration protocol.

Guarantees open-science research integrity.

ACTION ITEM 3

Collaborative Mentorship

Joint statistical review, pedagogical interpretation of findings, and co-authorship through publication submission.

Targeting Scopus Q1 co-authorship.

"The infrastructure is active, the database is live, the methodology is sound, and the research gap addresses a critical global challenge."

Let us execute this empirical research into high-impact IEEE and Scopus Q1 publications for the institution.

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